

AABAN REHMAN

.NET AI Engineer · RAG & LLM Systems · ASP.NET Core · Microservices · Azure

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PROFILE

.NET Engineer with 1.5+ years shipping enterprise microservices in ASP.NET Core, C#, EF Core, SQL Server, Docker, and Azure, now specializing in AI Engineering. I build production-grade RAG pipelines, LLM-orchestrated agents, and vector search systems in .NET. Personal projects include an agentic proposal-generation engine and a semantic Islamic knowledge assistant backed by pgvector — both built with a near-zero hallucination tolerance as a hard design constraint. Microsoft certified in Azure AI Fundamentals (AI-900) and Azure Fundamentals (AZ-900).

TECHNICAL SKILLS

AI / ML	RAG Pipelines · LLM Integration (Llama 3.3-70B) · Semantic Embeddings · Vector Search · pgvector · nv-embed-v1 (4096-dim) · NVIDIA Build API · Reranking API · Chunking & Retrieval · Agentic AI (Microsoft.Agents.AI 1.0) · Prompt Engineering · Tool Use / Function Calling
Backend	C# · .NET 8/9/10 · ASP.NET Core · EF Core · Dapper · Minimal APIs · Background Services · FluentValidation · Scalar UI · xUnit
Architecture	Microservices · DDD · Vertical Slice Architecture · Clean Architecture · CQRS · Event-Driven · Primary Constructor DI
Data	PostgreSQL · pgvector · SQL Server · MongoDB · Cosmos DB · Redis · Npgsql · Cursor-based Pagination
Cloud & DevOps	Azure (App Services, SQL, Key Vault, Service Bus) · Docker · GitHub Actions · CI/CD · AWS EC2
Frontend	Blazor · React · TypeScript · Bootstrap · Tailwind CSS
Security	OAuth2 · OpenID Connect · JWT · Azure AD SSO
Integrations	Stripe · PostGrid · USPS · Firebase · REST & Third-Party APIs

EXPERIENCE

- Full-Stack .NET Engineer, OwaSoft Technologies

Jul 2025 – Apr 2026 · Remote

 - Architected 10+ ASP.NET Core microservice projects using DDD and Vertical Slice Architecture, improving scalability and fault isolation across enterprise clients.
 - Designed RESTful APIs with OAuth2/JWT auth and OpenAPI/Swagger documentation, supporting concurrent mobile and web client teams.
 - Optimised MSSQL Server schemas and EF Core queries, reducing database latency by 35% on enterprise-scale data models.
 - Integrated Stripe, PostGrid, and USPS APIs, cutting third-party integration errors by 30% with zero downtime across thousands of monthly transactions.
 - Built ASP.NET Core background services for contract generation, invoicing, and push notifications — halving user-facing response times.
 - Led two Agile/Scrum projects as technical lead; coordinated mobile, web, and admin panel teams, cutting cross-team integration bugs by 25%.
- .NET Developer, Kartoa Technologies

Feb 2025 – May 2025 · Islamabad

 - Built full-stack features for WooqLaw.com legal-tech SaaS using Blazor and ASP.NET Core, increasing data retrieval speed by 22%.
 - Delivered 5+ reusable Blazor components and applied Clean Architecture principles across the codebase.

PROJECTS

- Islamic Knowledge Assistant

.NET 10 · PostgreSQL + pgvector · nv-embed-v1 · Llama 3.3-70B · ASP.NET Core

Semantic Q&A over the full Quran, Hadith corpus, and Ibn Kathir Tafsir — built with near-zero hallucination tolerance as a hard constraint.

 - Built a complete RAG pipeline in .NET 10: ETL for 6,236 Quranic ayahs (Tanzil XML), 60,000+ Hadiths (hadithapi.com REST with cursor pagination), and per-surah Ibn Kathir Tafsir (JSON) — all idempotent with ON CONFLICT upserts.
 - Implemented an embedding ingestion pipeline using nv-embed-v1 (4096-dim) via NVIDIA Build API into pgvector. Debugged production issues: EF change tracker memory bloat (ChangeTracker.Clear() per batch), RAM spikes from AsAsyncEnumerable (resolved with 50-row cursor batching), and an infinite-loop TextChunker.
 - Engineered the retrieval layer: SemanticSearchAsync, FindAliasesAsync, ApplyTranslationAsync, and a refactored Planner using Task.WhenAll for parallel alias resolution. Integrated NVIDIA Reranking API for top-k candidate reordering.
 - Every response enforces a mandatory citation and scholar consultation note — a hard architectural rule given the religious sensitivity of the domain.
- BidMaker Engine

.NET 9 · ASP.NET Core · Microsoft.Agents.AI 1.0 · Llama 3.3-70B · NVIDIA API · EF Core · SQL Server

Agentic AI system that generates tailored business proposals using LLM orchestration and parallel section generation.

- Designed a multi-tool AI agent with Microsoft.Agents.AI 1.0 and AIFunctionFactory — the LLM autonomously calls domain tools to gather company, project, and compliance data before drafting.
- Applied Task.WhenAll across independent proposal sections for parallel LLM calls, achieving a dramatic reduction in end-to-end generation time on multi-part documents.
- Used validated structured input for metadata extraction — avoiding hallucinated field values on business-critical proposal content.

NoorCards *.NET MAUI · SkiaSharp · Android · Offline-First*

- Offline Android app generating styled Arabic/English Islamic image cards with five custom SkiaSharp rendering templates.

EDUCATION

BS Software Engineering · International Islamic University Islamabad (IIUI) · Jan 2022 – Present (completion delayed)

CERTIFICATIONS

- Microsoft Azure AI Fundamentals (AI-900) ([view credential](#))
- Microsoft Azure Fundamentals (AZ-900) ([view credential](#))
- Microsoft Back-End Engineer Specialization ([view credential](#))
- Database Integration & Management ([view credential](#))
- Deployments & DevOps ([view credential](#))